Dynamic Links: Latent Link Generators in the Variational Free Energy

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Abstract

The exponential links $\Theta_{ij} = U_i e^{X_{ij}} U_j^{-1}$ of the gauge-covariant attention construction are promoted from parameters to latent random variables. The generators X_{ij} live on the edges of a depth-by-token lattice, receive a normalized prior, and are inferred together with the token states and routing labels, so that their dynamics are inference rather than a postulated mechanics. We write the joint model, derive its variational free energy in closed form for mean-field recognition, and box the resulting functional. Its link sector is an identity-anchored Gaussian prior plus a curvature energy on the elementary loops of the lattice. We show that every continuous conjugation-invariant curvature measure on the non-compact group $\mathrm{GL}(K)$ is blind to unipotent holonomy, and we use instead the Frobenius defect of the reference-coordinate holonomy, which separates every holonomy from the identity and equals Wilson's plaquette density on $O(K)$. Stationarity gives a link field equation whose source is a precision-weighted prediction error times the source state, the delta rule of fast-weight learning. Without the link prior the optimal transport is rank one and lies outside the exponential chart; generators are then stationary only at critical points of the exponential map, and the infimum over the chart is either approached at that singular transport or lies strictly above it, so the prior is necessary. Linearized, the equation is a screened, massive discrete Maxwell equation of Euclidean signature, in which the time-like loops of the lattice supply the kinetic term and its coefficient, the precision of the implied depth increment, acts as an inertial mass for the maximum a posteriori path. Inference itself is first-order relaxation. Finally, directed (causal) matter induces no action on the links, while integrating out undirected Gaussian matter induces the action $\frac{1}{2}\log\det J(X)$, bounded below, whose hopping expansion begins with Wilson's triangle action. All identities were checked in float64 at small size; nothing was trained.

1 Scope

The exponential link $\Theta_{ij} = U_i e^{X_{ij}} U_j^{-1}$ is taken here as the primary construction. VFE 3.0 implements its static, shared-generator form, selected by `content_link_chart="exponential"`, in which the generator is a single learned matrix $X = OT$, the same on every edge. This note asks what changes when each X_{ij} becomes a latent variable inferred from the context, so that the links evolve. The physics analogy is lattice gauge theory, in which link variables carry a Boltzmann weight e^{-S} [1–3]. In the Euclidean setting that weight is a probability measure, so a variational free energy over latent links is the natural home for link dynamics; no separate Lagrangian is needed. The kinetic and curvature terms then appear as parts of $-\log$ of a declared prior over link fields, and the remaining question is which prior. Sections 3 and 8 address that question from two sides: what a curvature energy on a non-compact group must look like, and what curvature energy matter induces when it is integrated out.

The scope is fixed throughout. The generative model is the layered causal model of Section 4, with depth as the time axis. Recognition is mean field over token states, routing labels and links. The checks of Appendix A are float64 computations at $K \leq 4$ with a few tokens; they ground identities and code paths and do not set the direction of the theory, and no statement here is a training result.

2 The depth-by-token lattice

Tokens are indexed by $i \in \{0, \dots, N-1\}$ with causal support $\mathcal{S}_i = \{j : j \leq i\}$, and layers by $\ell \in \{0, \dots, L\}$. Each token carries a frame $U_i \in \mathrm{GL}(K)$, built once per forward pass and shared across layers as in the current architecture. The latent state of token i at layer ℓ is written in reference coordinates, $\zeta_i^\ell \in \mathbb{R}^K$, with local coordinates $U_i \zeta_i^\ell$. A passive frame change $U_i \mapsto g_i U_i$ leaves every reference-coordinate quantity unchanged; this is the gauge convention used throughout.

A link edge connects the vertex (j, ℓ) to $(i, \ell+1)$ for every $j \in \mathcal{S}_i$. The edge carries a generator $X_{ij}^\ell \in \mathfrak{gl}(K)$ in reference coordinates, its group element $G_{ij}^\ell = e^{X_{ij}^\ell}$, and the local link

$$\Theta_{ij}^\ell = U_i G_{ij}^\ell U_j^{-1}, \qquad \Theta_{ij}^\ell \mapsto g_i \Theta_{ij}^\ell g_j^{-1}. \tag{1}$$

The self edge $(i, \ell) \to (i, \ell+1)$ is time-like and the edges with $j < i$ are cross-token. Two-step routes from (k, ℓ) to $(i, \ell+2)$ pass through $(j, \ell+1)$ with $k \leq j \leq i$. We compare each route with the one that first stays at k, and define for $0 \leq \ell \leq L-2$

$$W_{ijk}^\ell = (G_{ik}^{\ell+1} G_{kk}^\ell)^{-1} G_{ij}^{\ell+1} G_{jk}^\ell, \qquad k < j \leq i. \tag{2}$$

The loop is trivial for $j = k$. For $j = i$ it is time-like: it compares moving at layer ℓ and staying at $\ell+1$ with staying first and moving second. For $k < j < i$ it is space-like. Any two routes through j and j' differ by $(W_{ij'k}^\ell)^{-1} W_{ijk}^\ell$, so the loops (2) generate all route comparisons between layers ℓ and $\ell+2$. Because X is written in reference coordinates, W_{ijk}^ℓ itself, and not only its conjugacy class, is invariant under the passive action; the local holonomy is $U_k W_{ijk}^\ell U_k^{-1}$. When the links do not depend on the layer and the time-like links are trivial, W_{ijk}^ℓ reduces to the inverse-edge triangle holonomy $F_{ijk} = G_{ik}^{-1} G_{ij} G_{jk}$ of a layer-independent link field.

3 Measuring curvature on a non-compact group

A curvature energy must vanish on flat configurations and be positive otherwise. On compact groups the conjugation-invariant functions do this; on $\mathrm{GL}(K)$ they cannot.

Proposition 3.1 (Class functions are blind to unipotent holonomy)**.** *Let $K \geq 2$ and let $f : \mathrm{GL}(K, \mathbb{R}) \to \mathbb{R}$ be continuous and constant on conjugacy classes. Then $f(u) = f(I)$ for every unipotent u.*

Proof. Write $u = RJR^{-1}$ with J in Jordan form, so that $J = I + N$ with N nilpotent and supported on the superdiagonal. Let $D_t = \mathrm{diag}(1, t^{-1}, t^{-2}, \dots, t^{-(K-1)})$. Then $(D_t J D_t^{-1})_{p,p+1} = t J_{p,p+1}$, so $D_t J D_t^{-1} \to I$ as $t \to 0$, and $f(u) = f(D_t R^{-1} u R D_t^{-1}) \to f(I)$. □

The same scaling applied to the real Jordan form shows that such an f depends only on the semisimple part of its argument, that is, on the eigenvalues. Hence the eigenvalue density $\sum_k |\lambda_k(W) - 1|$, the spectral curvature $(\frac{1}{K} \sum_k |\mathrm{Log} \lambda_k|^2)^{1/2}$ of the implemented routing-holonomy diagnostic, and every other eigenvalue statistic vanish on the shear $W = I + E_{12} \neq I$ (Check 1). In particular, the bounded eigenvalue density of the multi-agent code does not vanish only at $W = I$.

Definition 3.2 (Frobenius defect)**.** For a reference-coordinate holonomy W, set $\Phi(W) = \frac{1}{2}\|W - I\|_F^2$.

Proposition 3.3 (Properties of the Frobenius defect)**.** *The defect is nonnegative and vanishes exactly at $W = I$. On $O(K)$ it equals Wilson's plaquette density, $\Phi(W) = K - \operatorname{tr} W$. It is invariant under the passive gauge action. Under the reparameterization $U_k \mapsto U_k h_k^{-1}$, $G_{ab}^{\ell} \mapsto h_a G_{ab}^{\ell} h_b^{-1}$, which leaves every local link unchanged, the holonomy transforms as $W_{ijk}^{\ell} \mapsto h_k W_{ijk}^{\ell} h_k^{-1}$, and $\Phi(h_k W h_k^{-1}) = \Phi(W)$ for every $W \in \mathrm{GL}(K)$ exactly when h_k is a nonzero multiple of an orthogonal matrix. For small generators,*

$$\Phi(W_{ijk}^{\ell}) = \tfrac{1}{2}\|(\delta X)_{ijk}^{\ell}\|_F^2 + O(\|X\|^3), \qquad (\delta X)_{ijk}^{\ell} = X_{ij}^{\ell+1} + X_{jk}^{\ell} - X_{ik}^{\ell+1} - X_{kk}^{\ell}. \tag{3}$$

For a time-like loop, $j = i$, this is $(\delta X)_{iik}^{\ell} = -\big[(X_{ik}^{\ell+1} - X_{ik}^{\ell}) - (X_{ii}^{\ell+1} - X_{kk}^{\ell})\big]$, a discrete electric field: the depth difference of a cross-token link corrected by the time-like links at its ends.

Proof. Nonnegativity and the zero set are properties of the norm. For orthogonal W, $\|W - I\|_F^2 = \operatorname{tr}(W^\top W) - 2\operatorname{tr} W + K = 2(K - \operatorname{tr} W)$. Passive invariance holds because W is reference data. Under the reparameterization each two-step route from (k, ℓ) to $(i, \ell + 2)$ becomes $h_i(\cdot)h_k^{-1}$, so $W \mapsto h_k W h_k^{-1}$. If $h_k = cO$ with $c \neq 0$ and O orthogonal, then $h_k(W - I)h_k^{-1} = O(W - I)O^\top$ has the Frobenius norm of $W - I$. Conversely, for $u, v \neq 0$ and small $t \neq 0$ the matrix $W = I + tuv^\top$ is invertible, and $\|h_k(W - I)h_k^{-1}\|_F = |t||h_k u||h_k^{-\top} v|$, so invariance gives $|h_k u|/|u| = |v|/|h_k^{-\top} v|$ for all nonzero u and v. Both sides are then one constant $c > 0$, so $|h_k u| = c|u|$ for every u and h_k/c is orthogonal. The expansion follows from $W = (I - X_{ik}^{\ell+1} - X_{kk}^{\ell})(I + X_{ij}^{\ell+1} + X_{jk}^{\ell}) + O(\|X\|^2)$. □

The Frobenius defect thus uses one piece of structure that class functions do not: the reference metric. That metric reduces the measurement of curvature from $\mathrm{GL}(K)$ to its compact subgroup $O(K)$, where class functions would suffice: the reparameterizations that preserve Φ are the nonzero multiples of orthogonal matrices, and the scalar factor acts trivially on W. In local coordinates the metric is carried by the frame field. The same metric reappears below as the isotropic reference noise of the transport kernel and as the Frobenius anchor of the link prior; all three must be the same metric for the model to have the residual $O(K)$ symmetry of Proposition 3.3. A uniform scalar reparameterization leaves every link and the anchor unchanged but rescales the reference states, and the transport noise and the token priors fix that scale. The version of Proposition 3.3 dated October 3, 2026 stated invariance exactly for orthogonal h_k, which omitted the scalar multiples.

4 The generative model with latent links

Definition 4.1 (Latent-link model)**.** The joint density of observations o, states ζ, head labels J and links X is

$$p(o, \zeta, J, X) = p(X) \prod_i p(\zeta_i^0) \prod_{\ell=0}^{L-1} \prod_i \Big[\prod_h \pi_{i,J_i^{\ell,h}}^{\ell,h} \Big] \mathcal{N}(\zeta_i^{\ell+1}; m_i^{\ell}, \bar{Q}) \prod_i p(o_i \mid U_i \zeta_i^L), \tag{4}$$

with transport mean and link prior

$$m_i^{\ell} = \sum_h G_{i,J_i^{\ell,h}}^{\ell} P_h \zeta_{J_i^{\ell,h}}^{\ell}, \qquad p(X) = Z_X^{-1} e^{-S_X(X)}. \tag{5}$$

Here $p(\zeta_i^0) = \mathcal{N}(\xi_{x_i}^p, C_{x_i}^p)$ is the reference-coordinate prior of token type x_i, $\pi^{\ell,h}$ is the causal routing prior, $\bar{Q} \succ 0$ is the reference transport noise, and $P_h = T^{-1} D_h T$ are the head projectors of the tiled

heads, with T the invertible stacked sending map and D_h the coordinate projector onto the block of head h, so that $P_h P_g = \delta_{hg} P_h$ and $\sum_h P_h = I$. The link action is

$$S_X(X) = \sum_{\ell=0}^{L-1} \sum_i \sum_{j \in \mathcal{S}_i} \frac{\|X_{ij}^\ell - \bar{X}^\ell\|_F^2}{2\sigma^2} + \lambda \sum_{\ell=0}^{L-2} \sum_{k<j\le i} w_{ijk}^\ell \Phi(W_{ijk}^\ell), \qquad w_{ijk}^\ell = \bar{\pi}_{ij}^{\ell+1} \bar{\pi}_{jk}^\ell, \tag{6}$$

with $\bar{\pi}$ the head-averaged routing prior, an anchor $\bar{X}^\ell$, anchor width $\sigma > 0$ and curvature coupling $\lambda \ge 0$.

Each head carries the component $P_h \zeta_j$ that it reads from its own source and transports it through that source's link, so the link acts after the head reads. The path weights in (6) use the generative routing prior and never the recognition weights, in keeping with the modeling rule, adopted here, that no generative factor reads a recognition law. The model is properly normalized: $S_X \ge \|X - \bar{X}\|_F^2/(2\sigma^2)$, so Z_X is finite, and every other factor in (4) is a normalized conditional. All link-sector quantities are reference data, so the passive gauge action changes only the frames, which enter the joint only through the decoder.

Remark 4.2 (Static limit). If $\bar{X}^\ell = OT$ for every layer and $\sigma \to 0$, every link is pinned to the shared generator and the mean of m_i^ℓ under routing weights β is $e^{OT} T^{-1} B_{ij} T \xi_j$ summed over j, the exponential-chart message with shared generator OT (Check 10). The implemented `content_link_chart="exponential"` is therefore the zero-temperature limit of this model, and the dynamic links are the fluctuations around it. Per-head links $X_{ij}^{\ell,h}$ are a variant that this note does not pursue.

5 The variational free energy

Recognition is mean field,

$$q(\zeta, J, X) = \prod_{\ell,i} \mathcal{N}(\zeta_i^\ell; \xi_i^\ell, C_i^\ell) \prod_{\ell,i,h} \beta_i^{\ell,h}(J_i^{\ell,h}) q(X), \tag{7}$$

with $q(X)$ either a point mass at M, which gives maximum a posteriori links, or a Gaussian on $\mathfrak{gl}(K)^{\text{edges}}$. With $A = \bar{Q}^{-1}$, write the routed moments of the transport mean under q,

$$\bar{m}_i^\ell = \mathbb{E}_q[m_i^\ell], \qquad V_i^\ell = \mathrm{Cov}_q(m_i^\ell). \tag{8}$$

For a point mass $q(X)$, with $a_{ij}^h = G_{ij}^\ell P_h \xi_j^\ell$, $\bar{m}_i^h = \sum_j \beta_{ij}^{\ell,h} a_{ij}^h$, $c_{ij}^{hh} = \beta_{ij}^{\ell,h}$ and $c_{ij}^{hg} = \beta_{ij}^{\ell,h} \beta_{ij}^{\ell,g}$ for $h \ne g$,

$$V_i^\ell = \sum_h \sum_j \beta_{ij}^{\ell,h} (a_{ij}^h - \bar{m}_i^h)(a_{ij}^h - \bar{m}_i^h)^\top + \sum_{h,g} \sum_j c_{ij}^{hg} G_{ij}^\ell P_h C_j^\ell P_g^\top G_{ij}^{\ell\top}, \tag{9}$$

the moment-matched predictive covariance of the current implementation without its floor.

Lemma 5.1 (Transport term)**.** *Under* (7)*,*

$$\mathbb{E}_q\big[-\log \mathcal{N}(\zeta_i^{\ell+1}; m_i^\ell, \bar{Q})\big] - H\big[q(\zeta_i^{\ell+1})\big] = D_{\mathrm{KL}}\big(q(\zeta_i^{\ell+1}) \| \mathcal{N}(\bar{m}_i^\ell, \bar{Q})\big) + \tfrac{1}{2} \operatorname{tr}\big(A V_i^\ell\big). \tag{10}$$

For one head and a point mass $q(X)$ the right side equals

$$\sum_j \beta_{ij}^\ell \Big[D_{\mathrm{KL}}\big(q(\zeta_i^{\ell+1}) \| \mathcal{N}(G_{ij}^\ell \xi_j^\ell, \bar{Q})\big) + \tfrac{1}{2} \operatorname{tr}\big(A G_{ij}^\ell C_j^\ell G_{ij}^{\ell\top}\big) \Big]. \tag{11}$$

Proof. Under mean field $\zeta_i^{\ell+1}$ is independent of m_i^ℓ, so $\mathbb{E}_q[(\zeta-m)(\zeta-m)^\top] = C_i^{\ell+1} + (\xi_i^{\ell+1} - \bar{m}_i^\ell)(\xi_i^{\ell+1} - \bar{m}_i^\ell)^\top + V_i^\ell$. Inserting this into the Gaussian cross entropy and subtracting the entropy $\frac{1}{2}\log\det(2\pi e C_i^{\ell+1})$ gives (10). For one head, $V_i^\ell = \sum_j \beta_{ij} G_{ij} C_j G_{ij}^\top + \sum_j \beta_{ij}(a_j - \bar{m})(a_j - \bar{m})^\top$, and for Gaussians with a common covariance $\sum_j \beta_j D_{\mathrm{KL}}(q\|\mathcal{N}(a_j, \bar{Q})) = D_{\mathrm{KL}}(q\|\mathcal{N}(\bar{m}, \bar{Q})) + \frac{1}{2}\sum_j \beta_j \|a_j - \bar{m}\|_A^2$, which gives (11). □

Equation (11) identifies the matter coupling: a routing-weighted alignment between the receiver's belief and each transported source, in which the source's uncertainty is pushed through the link and charged in noise units. It is the expected-likelihood (channel) form of the alignment energy rather than the ambient sum of divergences, because the joint (4) is a likelihood model. Collecting the factors of (4) gives the free energy.

$$\begin{aligned}\mathcal{F}[q] = &\sum_i \mathbb{E}_q\big[-\log p(o_i \mid U_i \zeta_i^L)\big] + \sum_i D_{\mathrm{KL}}\big(q(\zeta_i^0)\|p(\zeta_i^0)\big) \\ &+ \sum_{\ell=0}^{L-1}\sum_i \Big[D_{\mathrm{KL}}\big(q(\zeta_i^{\ell+1})\|\mathcal{N}(\bar{m}_i^\ell, \bar{Q})\big) + \tfrac{1}{2}\operatorname{tr}\big(\bar{Q}^{-1}V_i^\ell\big)\Big] + \sum_{\ell,i,h} D_{\mathrm{KL}}\big(\beta_i^{\ell,h}\|\pi_i^{\ell,h}\big) \\ &+ \mathbb{E}_{q(X)}\Big[\sum_{\ell,i,j\in\mathcal{S}_i} \frac{\|X_{ij}^\ell - \bar{X}^\ell\|_F^2}{2\sigma^2} + \lambda \sum_\ell \sum_{k<j\le i} \bar{\pi}_{ij}^{\ell+1}\bar{\pi}_{jk}^\ell \tfrac{1}{2}\big\|W_{ijk}^\ell - I\big\|_F^2\Big] - H\big[q(X)\big] + \log Z_X\end{aligned} \tag{12}$$

The first line is the evidence and encoding sector, the second the matter sector of Lemma 5.1 together with the routing sector, and the third the link sector, $D_{\mathrm{KL}}(q(X)\|p(X))$, with the anchor acting as a mass term and the loop energy as the curvature term. By Jensen's inequality $\mathcal{F}[q] \ge -\log p(o)$ for every q, so (12) is an exact variational bound for the declared model. For a point mass $q(X) = \delta_M$ the entropy and $\log Z_X$ are constants and the third line is $S_X(M)$. Because the links enter $\bar{m}$ and V, every sector of the second line couples to them.

6 The link field equation

Write $L(X, E) = \int_0^1 e^{sX} E e^{(1-s)X} \mathrm{d}s$ for the Fréchet derivative of the matrix exponential [7]. Its Frobenius adjoint is $L(X^\top, \cdot)$, so $\partial f(e^X)/\partial X = L(X^\top, \partial f/\partial G)$.

Proposition 6.1 (Field equation)**.** *For point-mass links, $\mathcal{F}$ is stationary in X_{ij}^ℓ exactly when*

$$\frac{\partial S_X}{\partial X_{ij}^\ell} = L(X_{ij}^{\ell\top}, \mathcal{J}_{ij}^\ell), \qquad \mathcal{J}_{ij}^\ell = \sum_h \beta_{ij}^{\ell,h} A \mathbb{E}_q[(\zeta_i^{\ell+1} - m_i^\ell)\zeta_j^{\ell\top} \mid J_i^{\ell,h} = j] P_h^\top. \tag{13}$$

For one head the matter current is

$$\mathcal{J}_{ij}^\ell = \beta_{ij}^\ell A\Big[(\xi_i^{\ell+1} - G_{ij}^\ell \xi_j^\ell)\xi_j^{\ell\top} - G_{ij}^\ell C_j^\ell\Big]. \tag{14}$$

Proof. Only the transport term of receiver i at layer ℓ and the link action depend on X_{ij}^ℓ. By Lemma 5.1 the transport term is $\frac{1}{2}\mathbb{E}_q\|\zeta_i^{\ell+1} - m_i^\ell\|_A^2$ up to X-free terms. Since m_i^ℓ contains $G_{ij}^\ell P_h \zeta_j^\ell$ exactly on the event $J_i^{\ell,h} = j$, its derivative in G_{ij}^ℓ is $-\mathcal{J}_{ij}^\ell$, and the chain rule through e^X gives (13). For one head, $\mathbb{E}[(\zeta' - m)\zeta_j^\top \mid J = j] = \xi'\xi_j^\top - G_{ij}(C_j + \xi_j\xi_j^\top)$, which is (14). □

The current is a precision-weighted prediction error times the source state, minus a term from the source's uncertainty. It is the delta rule, the update of fast-weight memories [11], here derived from a free energy and filtered through the derivative of the exponential chart. The left side of (13) is the response of the link action. The equation has the form of a field equation with a source: link stress equals matter current.

Corollary 6.2 (The link prior is necessary)**.** *Let $S_X = 0$, one head, $K \geq 2$ and $\beta_{ij}^\ell > 0$, and write $\xi' = \xi_i^{\ell+1}$, $S_j^\ell = C_j^\ell + \xi_j^\ell \xi_j^{\ell\top}$ and $\mathcal{T}_{ij}(G) = \frac{1}{2}\beta_{ij}^\ell \mathbb{E}_q \|\zeta_i^{\ell+1} - G\zeta_j^\ell\|_A^2$ for real $K \times K$ matrices G. Up to an X-free constant, $\mathcal{F}$ depends on X_{ij}^ℓ through $\mathcal{T}_{ij}(e^{X_{ij}^\ell})$, and:*

(i) $\mathcal{T}_{ij}$ is a strictly convex quadratic whose only stationary point is $G^\star = \xi' \xi_j^{\ell\top} (S_j^\ell)^{-1}$, of rank at most one and therefore outside the exponential chart, since $\det e^X = e^{\operatorname{tr} X} > 0$.

(ii) A generator X_{ij}^ℓ is stationary only where $L(X_{ij}^{\ell\top}, \cdot)$ is singular, at a critical point of the exponential map. Such generators exist: for $K = 2$, $\xi' = \xi_j^\ell = e_1$ and $C_j^\ell = A = I$, the generator $X = -(\log 3)I + 2\pi \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is stationary with link $\frac{1}{3}I \neq G^\star$.

(iii) If $\xi_j^{\ell\top}(C_j^\ell)^{-1}\xi' \geq 0$, the infimum of $\mathcal{T}_{ij}$ over the exponential chart equals $\mathcal{T}_{ij}(G^\star)$, is not attained, and is approached only along links converging to $G^\star$.

(iv) If $\xi_j^{\ell\top}(C_j^\ell)^{-1}\xi' < 0$, no sequence of chart links converges to $G^\star$, and the infimum over the chart lies strictly above $\mathcal{T}_{ij}(G^\star)$. For $K = 2$ and $K = 3$ it is not attained; this last statement is not proved here.

Proof. By Lemma 5.1 the X_{ij}^ℓ-dependent part of $\mathcal{F}$ is $\mathcal{T}_{ij}(e^{X_{ij}^\ell})$ up to a constant. The gradient of $\mathcal{T}_{ij}$ is $\beta_{ij}^\ell A(GS_j^\ell - \xi'\xi_j^{\ell\top})$ and its Hessian $E \mapsto \beta_{ij}^\ell AES_j^\ell$ is positive definite, as in the proof of Proposition 6.3; indeed $\mathcal{T}_{ij}(G) - \mathcal{T}_{ij}(G^\star) = \frac{1}{2}\beta_{ij}^\ell \|A^{1/2}(G - G^\star)(S_j^\ell)^{1/2}\|_F^2$. This gives (i). By Proposition 6.1, X is stationary exactly when $L(X^\top, \nabla\mathcal{T}_{ij}(e^X)) = 0$, and $\nabla\mathcal{T}_{ij}(e^X) \neq 0$ because $e^X \neq G^\star$, so $L(X^\top, \cdot)$ is singular. In the example, $\nabla\mathcal{T}_{ij}(\frac{1}{3}I) = \frac{1}{3}\beta_{ij}^\ell \operatorname{diag}(-1, 1)$, and $\operatorname{diag}(-1, 1)$ anticommutes with the rotation generator J, and $L(X^\top, \operatorname{diag}(-1, 1)) = \frac{1}{3}\operatorname{diag}(-1, 1)\int_0^1 e^{2\pi(1-2s)J}\mathrm{d}s = 0$. For (iii) and (iv), write $\tau = \operatorname{tr} G^\star = \xi_j^{\ell\top}(S_j^\ell)^{-1}\xi'$, which by the Sherman-Morrison formula is $\xi_j^{\ell\top}(C_j^\ell)^{-1}\xi'$ divided by $1 + \xi_j^{\ell\top}(C_j^\ell)^{-1}\xi_j^\ell$. The eigenvalues of $G^\star$ are τ and 0 with multiplicity $K - 1$. A real matrix is the exponential of a real matrix exactly when it is invertible and every Jordan block belonging to a negative eigenvalue occurs an even number of times [6, Theorem 1]. If $\tau \geq 0$, then $G^\star + \varepsilon I$ has only positive eigenvalues for $\varepsilon > 0$, so it is a chart link converging to $G^\star$, and by the quadratic identity every sequence of links whose energy tends to $\mathcal{T}_{ij}(G^\star)$ converges to $G^\star$. If $\tau < 0$, it is a simple eigenvalue of $G^\star$, so every real matrix near $G^\star$ has a simple negative eigenvalue near τ, a single Jordan block of size one; such a matrix is no chart link. A neighborhood of $G^\star$ therefore contains no chart link, and outside it the quadratic identity keeps $\mathcal{T}_{ij}$ above $\mathcal{T}_{ij}(G^\star)$ by a positive amount. Non-attainment for $K = 2, 3$ requires a perturbation argument that covers every chart link at $K \leq 3$; that argument is not given here, so this part of (iv) is not established in this note. □

The version of this corollary dated October 3, 2026 stated that the infimum is always approached at the singular boundary. That holds only in case (iii). In case (iv), which a single context reaches on an open set of states, the chart infimum lies at strictly higher energy, and whether it is attained for $K \geq 4$, at a link with a negative eigenvalue whose Jordan blocks have size at least two, is open. In every case the stationary generators of the unregularized problem, where they exist, sit at critical points of the exponential map.

A single context supplies one source state per edge, which determines only a rank-one regression. Averaged over a population of contexts, the same normal equation gives the population regression link $C_{ab}C_{bb}^{-1}$, whose curvature measures conditional dependence; that curvature statement is not derived in this note. The anchor σ is what makes the per-context problem well posed, and its mean $\bar{X}$ is the learned, slow part of the link.

Proposition 6.3 (Linearized field equation)**.** *Take one head, a flat anchor $\bar{X}^\ell = 0$, small links and small sources. To first order the field equation is*

$$\frac{1}{\sigma^2}X_e + \lambda(\delta^\top w \delta X)_e + \beta_e A X_e S_{j(e)} = \beta_e A[(\xi_i^{\ell+1} - \xi_j^\ell)\xi_j^{\ell\top} - C_j^\ell], \tag{15}$$

for every edge $e = (i \leftarrow j)$ at layer ℓ, where δ is the loop map (3) and w the diagonal of loop weights. The operator on the left is self-adjoint in the Frobenius inner product and bounded below by σ^{-2}, so the linearized links are unique.

Proof. At $X = 0$, $L(0, \cdot)$ is the identity and $G = I + X + O(X^2)$, so (14) is $\beta A[(\xi' - \xi_j)\xi_j^\top - C_j - XS_j]$ plus terms of second order. By (3) the loop energy is $\frac{\lambda}{2}\sum w\|\delta X\|_F^2 + O(X^3)$, whose gradient is $\lambda\delta^\top w\delta X$. The map $X \mapsto AXS$ is self-adjoint and nonnegative because $\langle Y, AXS\rangle = \mathrm{tr}(Y^\top AXS) = \langle AYS, X\rangle$ and $\langle X, AXS\rangle = \|A^{1/2}XS^{1/2}\|_F^2$, and $\delta^\top w\delta$ is nonnegative. □

Read as physics, (15) is a static massive Maxwell (Proca) equation on the depth-by-token complex. The anchor is a bare mass, the loop term is the curl-curl operator, and the matter term βAXS is an additional, state-dependent mass, analogous to the screening a charged medium gives the photon. The source is the prediction error times the source state. Through the time-like loops, the loop term contains $\frac{\lambda}{2}w_{iik}^\ell\|(X_{ik}^{\ell+1} - X_{ik}^\ell) - (X_{ii}^{\ell+1} - X_{kk}^\ell)\|_F^2$, the square of a discrete electric field. The kinetic term is therefore not a separate postulate: it is the time-like part of the same curvature energy, exactly as the electric term arises from time-like plaquettes in lattice gauge theory [2], and its coefficient $\lambda\bar{\pi}_{ii}^{\ell+1}\bar{\pi}_{ik}^\ell$ is the curvature coupling times the prior probability of the route. The space-like loops give the magnetic part. The operator is positive, so the equation is elliptic: depth enters with the same sign as the token directions, the Euclidean signature (Check 6 compares (15) with the exact stationary links).

7 Dynamics

Two notions of dynamics follow from (12), and they should not be confused.

Inference dynamics. Within a forward pass the links relax toward the stationary point of $\mathcal{F}$. For point-mass links the natural preconditioner is the Fisher information of the transported law with respect to the generator, which for one head is

$$\mathcal{I}_{ij}^\ell[E, E'] = \beta_{ij}^\ell \,\mathrm{tr}\,(AL(X_{ij}^\ell, E)S_j^\ell L(X_{ij}^\ell, E')^\top), \tag{16}$$

the Gauss-Newton matrix of the transport term and the mean part of the Gaussian Fisher pullback of a link. The flow $\dot{M} = -\mathcal{I}(M)^{-1}\nabla_M\mathcal{F}$ is first order, with the Fisher information as friction and no inertia. For Gaussian $q(X) = \mathcal{N}(M, \Sigma_X)$ the corresponding natural-gradient updates are those of the Bayesian learning rule [8], $\Sigma_X^{-1} \leftarrow (1-\rho)\Sigma_X^{-1} + \rho\mathbb{E}_q[\nabla^2\ell]$ and $M \leftarrow M - \rho\Sigma_X\mathbb{E}_q[\nabla\ell]$, where ℓ is the X-dependent part of $\mathcal{F}$ before the entropy. In the Laplace approximation $\Sigma_X^{-1} = \mathcal{I} + \nabla^2 S_X$ at the stationary point, and the link entropy contributes an Occam term to (12).

Depth as time, and where inertia comes from. The maximum a posteriori link field over all layers solves (13) jointly in depth. The following statement makes the intuition of precision as inertial mass exact in the one place where it holds.

Proposition 7.1 (Inertia of the maximum a posteriori path)**.** *Let a path $X^0, \dots, X^L$ with fixed ends minimize $\sum_\ell \left[\frac{m}{2}\|X^{\ell+1} - X^\ell\|_F^2 + V(X^\ell)\right]$ with V differentiable. Then*

$$m(2X^\ell - X^{\ell+1} - X^{\ell-1}) + \nabla V(X^\ell) = 0, \tag{17}$$

the discretization of $m\ddot{X} = \nabla V$: Newton's equation with mass m in the potential $-V$.

Proof. Differentiate the objective in X^ℓ for $0 < \ell < L$. □

The quadratic increment term is $-\log$ of a Gaussian increment density with precision m, and in (12) it is supplied by the time-like loops with $m = \lambda \bar{\pi}_{ii}^{\ell+1} \bar{\pi}_{ik}^{\ell}$ when the time-like links are held at the identity. Precision is therefore the inertial mass of the smoothed path. Two qualifications follow from the sign. The equation is the Euclidean one, motion in the inverted potential, so the maximum a posteriori path between two wells is a monotone kink and never oscillates (Check 9). And the inertia belongs to the smoother, the joint optimization over depth; filtering layer by layer gives first-order dynamics. Real-time propagation, waves of the link field, would require an analytic continuation that no inference procedure supplies. Second-order dynamics in inference time can be obtained by declaring velocity states in the generative model, as in generalized coordinates of motion [10], which again ties the mass to a declared precision; the path-integral form of such increment priors is the Onsager-Machlup functional [9].

8 Where the curvature energy comes from

The loop energy in (6) is a declared prior. Integrating out the token states shows when such an energy is induced instead.

Proposition 8.1 (Directed matter induces no link action)**.** *In the model (4), $\int p(\zeta, J \mid X)\mathrm{d}\zeta\mathrm{d}J = 1$ for every X, so the prior marginal of the links is $p(X)$ itself.*

Proof. Every factor other than $p(X)$ and the decoder is a normalized conditional of a directed acyclic model, and the decoder integrates to one over o. □

In the causal model, curvature reaches the posterior only through evidence. The predictive law of $\zeta_i^{\ell+2}$ given ζ_k^ℓ is a mixture over routes j of Gaussians centered at $G_{ik}^{\ell+1} G_{kk}^\ell W_{ijk}^\ell \zeta_k^\ell$, whose components coincide exactly when the loops are trivial. Evidence generated by a single transport therefore favors flat links, while evidence that requires different routes to carry different content favors curvature.

An induced action appears for undirected matter. Let the states of one layer interact through pairwise consistency factors,

$$p(\zeta \mid X) \propto \exp\Big(-\tfrac{1}{2} \sum_v \zeta_v^\top \Lambda_0 \zeta_v - \tfrac{1}{2} \sum_{(i \leftarrow j)} \|\zeta_i - G_{ij}\zeta_j\|_\Lambda^2 \Big), \tag{18}$$

with $\Lambda_0, \Lambda \succ 0$, the energy-based reading of the alignment terms. Its precision matrix $J(X) = D(X) - \mathcal{A}(X)$ has diagonal blocks $D_v = \Lambda_0 + \sum_{v \leftarrow \cdot} \Lambda + \sum_{\cdot \leftarrow v} G^\top \Lambda G$ and off-diagonal blocks $\mathcal{A}_{ij} = \Lambda G_{ij}$, $\mathcal{A}_{ji} = G_{ij}^\top \Lambda$.

Proposition 8.2 (Induced action of undirected Gaussian matter)**.** *Let the joint density of states and links be proportional to $\tilde{p}(X)e^{-E(\zeta,X)}$, with E the exponent of (18) and $\tilde{p}$ a base prior, normalized as a whole and not conditionally in ζ. Then the link marginal is proportional to $\tilde{p}(X)e^{-S_{\mathrm{ind}}(X)}$, where*

$$S_{\mathrm{ind}}(X) = \tfrac{1}{2}\log\det J(X) \geq \tfrac{1}{2}\log\det(I \otimes \Lambda_0). \tag{19}$$

If the spectral radius of $D^{-1}\mathcal{A}$ is below one,

$$S_{\mathrm{ind}}(X) = \tfrac{1}{2}\log\det D - \tfrac{1}{2}\sum_{n\geq 1}\frac{1}{n}\operatorname{tr}\left[(D^{-1}\mathcal{A})^n\right], \tag{20}$$

and the n-th term is a sum over closed walks of length n of products of links, that is, of Wilson loops. For isotropic $\Lambda_0 = \lambda_0 I$, $\Lambda = \lambda_E I$ and orthogonal links, $D_v = d_v I$ does not depend on the links, the walks of length two contribute constants, and on a triangle

$$S_{\mathrm{ind}} = \mathit{const} + \frac{\lambda_E^3}{d_i d_j d_k}(K - \operatorname{tr} F_{ijk}) + O\Big(\frac{\lambda_E^5}{d^5}\Big), \tag{21}$$

which is $\lambda_E^3/(d_i d_j d_k)$ times the Frobenius defect $\Phi(F_{ijk})$.

Proof. Integrating $e^{-E(\zeta,X)}$ over ζ gives $(2\pi)^{nK/2}\det J(X)^{-1/2}$, so the link marginal is proportional to $\tilde{p}(X)\det J(X)^{-1/2}$. Had the states been given a conditionally normalized law, this factor would cancel, which is the content of Proposition 8.1. Each pairwise term is positive semidefinite, so $J \succeq I \otimes \Lambda_0$. The expansion is $\log\det(D - \mathcal{A}) = \log\det D + \log\det(I - D^{-1}\mathcal{A})$ with the logarithmic series. With orthogonal links, $G^\top \Lambda G = \lambda_E I$, so D is link independent, and a walk $i \to j \to i$ contributes $\lambda_E^2 \operatorname{tr}(GG^\top)/(d_i d_j)$, a constant. A closed walk around the triangle contributes $\lambda_E^3 \operatorname{tr} F/(d_i d_j d_k)$ in one orientation and $\lambda_E^3 \operatorname{tr} F^{-1} = \lambda_E^3 \operatorname{tr} F$ in the other; with three starting points and two orientations, the $n = 3$ term is $-\frac{1}{2}\cdot\frac{1}{3}\cdot 6\lambda_E^3 \operatorname{tr} F/(d_i d_j d_k)$. Proposition 3.3 identifies $K - \operatorname{tr} F$ with $\Phi(F)$. □

This is the Gaussian-matter version of the hopping-parameter expansion of lattice gauge theory [4] and of induced gauge actions [5]: integrating out matter generates the Wilson action at leading order, and the coupling is the cube of the hopping parameter. Check 8 confirms (21); the ratio of the exact induced action to the leading term is 1.15, 1.022, 1.003 and 1.0003 at $\lambda_0 = 3$, 10, 30 and 100. The induced action also settles the non-compact directions. With orthogonal links on a tree it is constant, but a symmetric generator on a single edge changes it through the diagonal blocks even without loops, so the non-compact directions acquire edge-local mass terms while the compact directions enter only through loops. The induced action is bounded below by (19) for every $\mathrm{GL}(K)$ configuration, and it responds to a unipotent holonomy that every class function misses (Check 8). The causal predictive route is directed, so Proposition 8.1 applies to it, and the loop energy of (6) is a declared prior whose form, the Frobenius defect with its Wilson limit, is the one undirected Gaussian matter would induce.

9 Relation to the implementation and next steps

The implemented exponential chart is the static limit of Remark 4.2. A first implementation of dynamic links would keep $\bar{X} = OT$ as the slow, learned anchor and infer per-context deviations $M - \bar{X}$ with a few Gauss-Newton steps preconditioned by (16), or with one solve of the linearized equation (15). The cost per step is one exponential per edge, $O(LN^2K^3)$ over all causal edges, which calls for restricted supports, low-rank deviations or amortization, in which a network predicts

$M - \bar{X}$ from the states; amortized deviations are a fast-weight program with the delta-rule source of Proposition 6.1. A grounding experiment at small size would check that inferred links reproduce the stationary equation, that the loop energy responds to unipotent holonomy, and that removing the anchor drives links toward the singular transport when $\xi_j^{\ell\top}(C_j^\ell)^{-1}\xi_i^{\ell+1} \geq 0$ and leaves them at strictly higher energy otherwise, as Corollary 6.2 predicts. Such checks validate the construction; they do not decide whether dynamic links help a language model at scale.

Several questions remain open. The values of λ, σ and the loop weights are modeling choices without a derivation here. The multi-head field equation couples edges through the receiver's mixture, and its linearization has cross-edge terms that this note does not analyze. Stationarity in the time-like links gives a constraint analogous to Gauss's law that remains to be worked out. Full Gaussian link posteriors cost $K^2 \times K^2$ per edge. Whether evidence-driven curvature improves prediction, and whether the undirected matter model should replace the declared loop prior, are empirical and modeling questions respectively.

A Numerical checks

The checks run in float64 with NumPy and SciPy. Check 1 evaluates the eigenvalue density and the spectral curvature on $I + E_{12}$ (both zero) and the Frobenius defect (0.5), and shows conjugates of the shear converging to I. Check 2 confirms $\Phi(Q) = K - \operatorname{tr} Q$ on 200 random orthogonal matrices to 1.8×10^{-15} and the small-field expansion (3) with relative error 10^{-3}, 3.4×10^{-4} and 3.8×10^{-5} at scales 10^{-1}, 10^{-2} and 10^{-3}. Check 3 compares the transport term (10) with a Monte Carlo estimate over 4×10^5 joint samples for two heads and three sources; the relative difference, 9.5×10^{-5}, is at the level of the sampling error, and the sampled mean and covariance of m match (8) and (9) to 1.3×10^{-3} and 1.4×10^{-2}. Check 4 confirms the one-head identity (11) exactly. Check 5 compares the gradient of Proposition 6.1 with central differences (3.0×10^{-10}) and confirms that the unregularized minimizer over all matrices has rank one. Check 6 compares the linearized links (15) with the exact stationary links of a four-token transition with triangle loops; the relative difference is 2.7×10^{-2}, 8.5×10^{-3} and 2.9×10^{-3} at source scales 10^{-1}, 3×10^{-2} and 10^{-2}, and the operator's smallest eigenvalue is 2.014 against the bound $\sigma^{-2} = 2$. Check 7 confirms the electric-loop expansion with relative error decreasing linearly in the scale. Check 8 confirms orthogonal gauge invariance of the induced action to machine precision, the hopping series to 10^{-15}, the leading Wilson coefficient through the ratios quoted in Section 8, the absence of an induced action for orthogonal links on a tree, its presence for a symmetric generator (0.237), its response to a unipotent link (0.045), and the lower bound over 500 random large GL(2) configurations. Check 9 confirms (17) at the computed minimizer to 7×10^{-7} and the monotone kink. Check 10 confirms the static limit of Remark 4.2 to 5.6×10^{-16}.

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